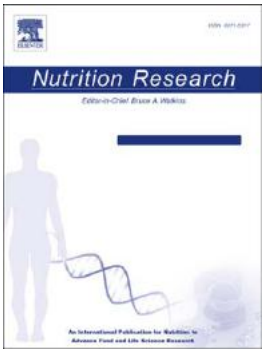

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Daily supplementation with mushroom (*Agaricusbisporus*) improves balance and working memory in aged rats

Nopporn Thangthaeng, Marshall G. Miller, Stacey M. Gomes, and Barbara Shukitt-Hale

USDA-ARS, Human Nutrition Research Center on Aging at Tufts University, Boston, MA

Correspondence to:

Barbara Shukitt-Hale, Ph.D.

USDA-ARS, HNRCAat Tufts University

711 Washington Street

Boston, MA 02111

Tel: 617-556-3118; Fax: 617-556-3299

E-Mail: barbara.shukitthale@ars.usda.gov

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Abstract

Decline in brain function during normal aging is partly due to the long-term effects of oxidative stress and inflammation. Several fruits and vegetables have been shown to possess antioxidant and anti-inflammatory properties. The present study investigated the effects of dietary mushroom intervention on mobility and memory in aged Fischer 344 rats. We hypothesized that daily supplementation of mushroom would have beneficial effects on behavioral outcomes in a dose-dependent manner. Rats were randomly assigned to receive a diet containing either 0, 0.5, 1, 2, or 5% lyophilized white button mushroom (*Agaricusbisporus*); after 8 weeks on the diet, a battery of behavioral tasks was given to assess balance, coordination, and cognition. Rats on the 2% or 5% mushroom-supplemented diet consumed more food, without gaining weight, than rats in the other diet groups. Rats in the 0.5% and 1% group stayed on a narrow beam longer, indicating an improvement in balance. Only rats on the 0.5% mushroom diet showed improved performance in a working memory version of the Morris water maze. When taken together, the most effective mushroom dose that produced improvements in both balance and working memory was 0.5%, equivalent to about 1.5 ounces of fresh mushrooms for humans. Therefore, the results suggest that the inclusion of mushroom in the daily diet may have beneficial effects on age-related deficits in cognitive and motor function.

Key words: Agaricusbisporus, white button mushroom, diet, aging, memory, balance

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1. Introduction

By the year 2030, approximately one-fifth of the US population will be older than 65 and will likely exhibit behavioral deficits resulting from numerous neuronal changes. These decrements can be expressed as alterations in both motor and cognitive behaviors [1]. The alterations in motor function may include decreases in balance, muscle strength and coordination, while cognitive deficits are seen on tasks that require spatial learning and memory [2]. Interestingly, several experiments carried out in our laboratory have indicated that it is possible to reverse these deficits during aging by utilizing berry or walnut supplementations [3].

Like many fruits and vegetables, mushrooms have been shown to arrest tumor growth [4], protect cells from oxidative damage [5,6], promote a healthy immune system [7] and lower cholesterol [8]. However, the effect of mushrooms on cognition has only recently been explored. In an epidemiologic study, Nurk and colleagues [9] reported a strong positive association between mushroom consumption and perceptual speed, executive function and semantic memory after examining the relationship between fruit and vegetable intake and cognitive performance among 2301 elderly males and females (70-74-year-old). These findings suggest beneficial effects of mushrooms on cognitive function in older adults; however, one caveat of this study was that mushroom eaters also reported higher fruit and vegetable intake. The first study to directly link mushroom intake to cognitive performance was a double-blind study by Mori and colleagues [10] where Yamabushitake (*Hericiumerinaceus*) mushrooms were administered to 50- to 80-year-old Japanese men and women who were diagnosed with mild cognitive impairment. Those who received Yamabushitake scored significantly better on tests of cognitive function than the

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control group, and this effect persisted for a short period after they stopped the treatment [10]. Moreover, mushrooms and mushroom extracts were recently shown to promote neurite outgrowth and ameliorate cognitive dysfunction in chemically-induced neurodegenerative mouse models [11].

Even though there are over 2,000 edible varieties of mushrooms, the white button mushroom, *Agaricusbisporus*, is the most popular worldwide [12]. Like other mushrooms, *A. bisporus* has been reported to have medicinal compounds with health benefits [13,14]. Wild-type and Alzheimer's model mice fed with vitamin D_2-enriched *A. bisporus* for 7 months exhibited improved learning and memory compared to those on control, no mushroom-supplemented, diet [15]. Therefore, *A. bisporus* may possess healthful nutrients that can be explored as a potential nutritional therapeutic to curtail brain aging. Moreover, to date, there are no studies that have examined the effect of mushrooms on normal brain aging or motor function. Thus, the purpose of this study was to assess the effects of graded dietary percentages of *A. bisporus* on age-related cognitive and motor deficits. The findings from this study will provide potential benefits of adding mushrooms to the diet on increased "healthspan" in aging.

2. Methods and Materials

2.1. Mushroom Diet

White button mushrooms (*Agaricusbisporus; Monterey Mushroom, Temple, PA*) were cleaned, stems were removed, and caps were sliced into equally sized pieces for lyophilization. The

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freeze-dried mushrooms were then pulverized and incorporated into a pelleted NIH-31 diet (Harlan Teklad; Madison, WI). Due to the absence of the empirical research highlighting the amount of mushroom intake on the both cognitive and motor function in aged rodents, five diets were formulated to cover a wide range of possible mushroom consumption in humans: control (0%), 0.5%, 1.0 %, 2.0%, or 5% mushroom. These doses are equivalent to ¼ (42.5 g), ½ (85g), 1 (170 g), or 2.5 (425 g) daily servings for humans. For the control diet, the amount of corn (yellow ground) was adjusted to compensate for the added amount of the mushrooms, as previously described [16] (Table 1).

2.2. Animals

Nineteen-month-old male Fischer 344 rats, acquired from the National Institute on Aging (NIA) rat colony (Taconic Farms), were maintained on a 12 h light/dark schedule in hanging wire cages with *ad libitum* food and water. After 2 weeks of acclimation, rats were weight matched and assigned to diet groups. Food intakes were recorded weekly and body weights were recorded every other week. During the course of the study, four rats in the control group, two rats in the 0.5% mushroom group, three rats in the 1.0% mushroom group, four rats in the 2.0% mushroom group, and one rat in the 5.0% mushroom group died or were euthanized due to excessive weight loss (more than 15% loss of body weight from the original, stabilized weight). Rats were monitored for signs of injury or disease in accordance with the HNRCA Institutional Animal Care and Use Committee (IACUC).

2.3. Behavioral Tests

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The battery of motor tests and the cognitive test selected for this study have been previously shown by our group to be age-sensitive [1]. For most of the measures, the decline in performance is observed as early as 12-15 months of age.

Previous work has shown that an N of 15 per group will give us sufficient power to find statistically significant ($p < 0.05$) differences on tests of behavior. Power calculations show that, for the MWM, a difference in means of 10 (e.g., 32 vs. 22 secs in latency to find the platform), when the standard deviation is 10, would yield a power of 78%, while for the rod walking test, a difference in means of 4.5 (e.g., 8.1 vs. 12.6 s in latency to fall), when the standard deviation is 4, would yield a power of 85%, using a sample size of 15..

2.3.1. Motor Testing

After 8 weeks of dietary intervention, rats underwent a battery of motor tests to assess for: 1) Balance: rats were placed on three planks of varying widths (13 mm, 25 mm, and 38 mm; counterbalanced) and a narrow rod (26 mm) horizontally suspended 23 cm above a thick foam-core pad and latency to fall was recorded (max 60 s); 2) Forelimb strength: rats were allowed to grasp a wire suspended 55 cm above a thick foam-core pad and latency to fall was recorded (max 60 s); 3) Overall limb strength: rats were placed on an inclined metal screen (60°) and latency to fall was recorded (max 600 s); 4) Fine motor coordination and stamina: rats were placed on an accelerating rotarod (Ugo Basile, Italy) consisting of a slowly accelerating (+2 rpm / 30 s; 20 rpm max) rotating dowel (7 cm diameter) and latency to fall was recorded (max 300 s).

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2.3.2. Cognitive Testing

To assess spatial learning and memory, rats completed a working memory version of the Morris water maze (wMWM) [17] at weeks 9-10 of dietary intervention. The test was given daily for 4 consecutive days, 2 sessions per day and 2 trials each session: a reference memory or acquisition trial (Trial 1) and a working memory or retrieval trial (Trial 2). During testing, rats were placed in a large water-filled pool (134 cm diameter) and allowed 120 s to escape onto a platform hidden 2 cm below the water surface. If the rat failed to escape within this time, it was guided to the platform. Once the rat reached the platform, it was allowed to remain there for 15 s (Trial 1). The rat was then returned to its home cage for 10 mins (inter-trial interval). Trial 2 used the same platform location and start position as Trial 1. The platform was moved to one of four locations, chosen to frustrate a number of non-place learning strategies that rats may adopt [18], at the beginning of each session. wMWM files were analyzed with image tracking software (HVS Image, Hampton, England).

2.4. Statistical Analyses

For each measure, between-subjects analysis of variance (ANOVA) models comparing the diet groups were performed using Systat (SPSS, Inc., Chicago, IL) to test for statistical significance at a level of $P < 0.05$. Days or trials, when appropriate, were included in the model as a within-subjects variable. To determine differences between the diet groups, Fisher's least significant difference (LSD) post-hoc analysis was performed. To analyze working memory, separate *t*-tests

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were conducted for each group between the Trial 1 and Trial 2 latencies and distances.

3. Results

3.1. Effect of graded mushroom supplementation on food intake and body weight

Food intake was monitored weekly starting two weeks after switching the rats onto their respective diet. A one-way ANOVA with repeated measures of the weekly food intake data revealed a significant main effect of diet and a significant interaction between diet and time (P's < 0.05; Figure 1A). Post-hoc testing showed that rats in the 2% mushroom diet group consumed significantly more than the control group ($P < 0.05$), while both of the higher-percentage mushroom-supplemented diet groups (2% and 5%) ate more than the two lower-percentage groups (Figure 1A). This difference was due to the fact that rats in the 2% and 5% mushroom groups initially ate more, but by the end of 8 weeks, rats in all diet groups consumed approximately similar amount of food. Interestingly, this increased food intake did not translate into weight gain (Figure 1B). A one-way ANOVA with repeated measures of the body weight over the course of the study revealed a significant main effect of time ($P < 0.05$), with no significant effects of diet. In general, rats in all diet groups lost weight slightly over the course of this study.

3.2. Effect of graded mushroom supplementation on motor function

Rats fed 0.5% and 1% mushroom performed significantly ($P < 0.05$) better on the large plank test

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compared to the controls (Figure 2). In other words, these 2 lower-percentage mushroom animals were able to stay on the large plank significantly longer, showing improved balance. There were no differences between the groups on either the small or medium planks. Additionally, rats in all groups performed similarly in wire suspension, inclined metal screen, rotarod and narrow rod balance tests (data not shown), suggesting that mushroom supplementation did not affect limb strength, fine motor coordination or stamina.

3.3. Effect of graded mushroom supplementation on working memory water maze performance

One-way ANOVAs with days of training as the repeated measure revealed a significant main effect of training days, but no significant different between the diet groups, for either Trial 1 or Trial 2. Overall, all the rats, except for those in 5% mushroom group, performed better over the four-days of training for both Trial 1 and Trial 2 (i.e., reductions in both latency and path length). In order to minimize the learning aspect of this "working memory" water maze test, averages of the latencies and distances across all four consecutive training days for each group were calculated. The effect of mushroom supplementation on working memory was determined by performing t-tests between the two trial (Trial 1 and Trial 2) latencies and distances for each group. When all days were analyzed, the 0.5% mushroom group was the only group showing a significant reduction in the latencies ($P < 0.05$) between Trial 1 and Trial 2 (Figure 3A). This result demonstrated that the rats fed with 0.5% mushroom were able to learn the platform location within one trial and retain the information after the 10 min inter-trial interval. Interestingly, a higher percentage of mushroom supplementation (1%-5%) appeared to hinder Trial 2 performance (Figure 3A). However, only the 2% mushroom group's latencies were

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significantly higher than the controls ($P < 0.05$). Data were similar for the distance parameter (Figure 3B). Differences observed between groups in latency were not due to swim speed as there were no differences in speed between groups (data not shown).

4. Discussion

This study examined the effect of short-term *A. bisporus*(white button mushroom) supplementation on age-related declines in motor and cognitive function in rats. The three major findings were: 1) no significant weight gain was observed, despite increased food intake; 2) dietary supplementation with *A. bisporus* can improve motor performance, specifically balance; and 3) small amounts of dietary supplementation with mushroom can improve cognition, while higher amounts may have negative effects, confirming our original hypothesis.

Rats fed with 2% and 5% mushroom diets consumed significantly more food, on average, than rats in the other diet groups without gaining more weight than rats in other diet groups. A small amount of yellow corn was removed from the diet in order to compensate for the changes in volume displaced by the addition of the lyophilized mushroom [16]. Since mushrooms are less calorie dense (22 calories/100 g) [19]than yellow corn (86 calories/100 g) [20], it is possible that rats in the 2% and 5% mushroom diet groups consumed slightly more to compensate for calories displaced from their diet by the increasing volume of lyophilized mushroom. Also, other mushrooms and their extracts have been proposed to have anti-obesity properties by regulating adipocyte differentiation [21,22]. Since *A. bisporus* possesses many of the same medicinal properties as the other mushrooms, it is possible that *A. bisporus* also contains these anti-obesity

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compounds which can only reach effective doses at higher amounts of mushroom. This finding is important because it demonstrates that mushroom can be added to the daily diet with negligible impact to body weight.

After eight weeks on the diets, aged rats completed a variety of motor behavior tests. While there were no differences in strength or coordination, rats fed with either a 0.5% or 1% mushroom diet were able to remain on the large plank significantly longer than controls, thus indicating an improvement in balance. Although it is not known as to why dietary daily intake of mushroom specifically improved balance, daily consumption of some vegetables, fruits, and nuts have been shown to improve performance on specific motor task(s), and not necessarily in a dose-dependent manner. For example, in a recent study by our laboratory, using the same model, in rats fed with one of five coffee-supplemented diets (0, 0.165, 0.275, 0.55, and 0.825%) for 8 weeks, showed that only the 0.55% group performed better on the rotarod, a test for coordination, as compared to control [23]. Similarly, a study by Willis and colleagues [24], where 19-month-old Fischer 344 rats were fed with either a control diet, 2%, 6% or 9% walnut supplemented diet for 8 weeks, reported the different doses of walnut had different effects on the performance on motor tests when compared to control. Rats fed with 2% walnut showed an improvement on rod walking and 6% walnut improved performance on medium plank; while, 9% walnut showed no improvement on any motor tasks. It is unclear as to why dietary daily intake of mushroom specifically improves balance; therefore, more research is needed to address this observation. Because falls and fall-related injuries are a major concern among older adults and a primary initiating factor for the transition from independent to assisted living, regular dietary supplementation with mushroom may improve motor control and delay the transition

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away from independent living.

In addition to motor behavior tests, aged rats also completed a working-memory version of the MWM. Rats fed with 0.5% mushroom were able to remember the platform location better than the controls by demonstrating one-trial learning. This result indicated an improvement in spatial working memory. Surprisingly, higher amounts of mushroom supplementation, particularly 2%, did not yield better cognitive performance. This finding is in contrast with the study by Nurk and colleagues, where a strong positive association between mushroom consumption and perceptual speed, executive function and semantic memory was observed in older adults [9]. Since those who identified as high mushroom eaters also reported higher fruit and vegetable consumption, the discrepancy between the studies could be due to other nutrients found in fruits and vegetables, or, perhaps, fruit and vegetable eaters have healthier living habits than non-fruit and vegetable eaters.

The negative effect of a higher percentage of mushroom supplementation on cognition also contradicts the findings reported by Bennett and colleagues [15]where 5% vitamin D_2-enriched *A. bisporus*-supplemented diet initiated at 2-months of age for 7-months, in both wild-type mouse and Alzheimer's mouse model, significantly improved the performance on a standard MWM test. The differences between the outcomes from these two studies could be due to the species of the animals used (rats vs. mice), the age of the implementation (19-month-old vs. 2-month-old), the cognitive test used (working memory version vs. the standard MWM which is more a test of learning), or the length of supplementation (8 weeks vs. 7 months). The significance of these factors on the effect of dietary mushroom supplementation on cognition

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will need to be investigated further.

Additionally, unlike Bennett and colleagues [15], *A. bisporus* used in this study was not treated to enrich vitamin D content. Vitamin D and its metabolites have been associated with lower risks of Alzheimer's [25], mild cognitive impairment [26], non-Alzheimer's dementias [27] and poorer cognitive performance [25] in older adults. As suggested by the authors, it is possible that the beneficial effects seen in Bennett et al. could be due to the supplementation of vitamin D_2 or the interaction between vitamin D_2 and other bioactive compound(s) in the mushroom [15].

One limitation of this study is the lack of young control group. Although the addition of young controls could attest to the age-sensitivity of the behavioral measures, previous studies by our laboratory have shown that all of the behavior tests conducted in this study were able to detect age-related declines in motor and cognitive function[1,28]. Moreover, we believed that the purpose of this study, to examine the effects of daily dietary inclusion of mushroom on cognitive and motor function in aged animals, was efficiently addressed by comparing the mushroom-fed groups to the age-matched control group. Additionally, including young rats masks the treatment effects as they are not usually improved by our dietary supplements, because they are not impaired.

In conclusion, the results from this study revealed that dietary supplementation with *A. bisporus* may have domain-specific effects on the normal aging brain. This study was the first to provide evidence suggesting that consumption of *A. bisporus* has beneficial effects on motor function. The most effective mushroom dose was 0.5%, in that improvements in both motor and cognitive

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function were observed. The findings from this study provide evidence suggesting potential health benefits of mushroom in older adults, albeit the beneficial effects observed were small, but significant, and conducted in aged rodents. An inclusion of approximately 42 g or 1.5 oz. or about 2 large white button mushrooms, an equivalent of 0.5%, in the daily diet may minimize the incidents of falls and extend the "healthspan" of older adults. However, further studies are still needed to investigate the mechanism(s) responsible for the beneficial behavioral effects observed and to explore the effects of mushroom on motor function in older adults.

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Conflicts of interest: None

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Figure legends

Figure 1 Mushroom consumption had no effect on body weight in spite of increased food intake in rats fed higher percentages of mushroom content. The results shown represent average food intakes (A, grams/day; means ± SEM; n = 11-14/group) over the course of the study for the various diet groups: 0, 0.5, 1, 2, or 5% lyophilized white button mushroom (*A. bisporus*) and average weight (B, grams; means ± SEM; n = 11-14/group) over the course of the study for the various diet groups. Means with different letters are significantly different ($P < 0.05$).

Figure 2 Effect of mushroom diet on plank walking performance. The results shown represent average latency to fall (seconds; mean ± SEM; n = 11-14/group) from the large plank. * denotes significant differences as compared to control rats ($P < 0.05$).

Figure 3 Effect of mushroom diet on working memory water maze performance. The results shown represent average latency in seconds (A; mean ± SEM; n = 11-14/group) and distance in meters (B; mean ± SEM; n = 11-14/group) to find the hidden platform on days 1- 4 of testing. The asterisk indicates a difference (i.e., an improvement) between Trial 1 and Trial 2 performance (** = $P < 0.01$), meaning that working memory was improved. The # indicates a difference from the control group (# = $P < 0.05$).

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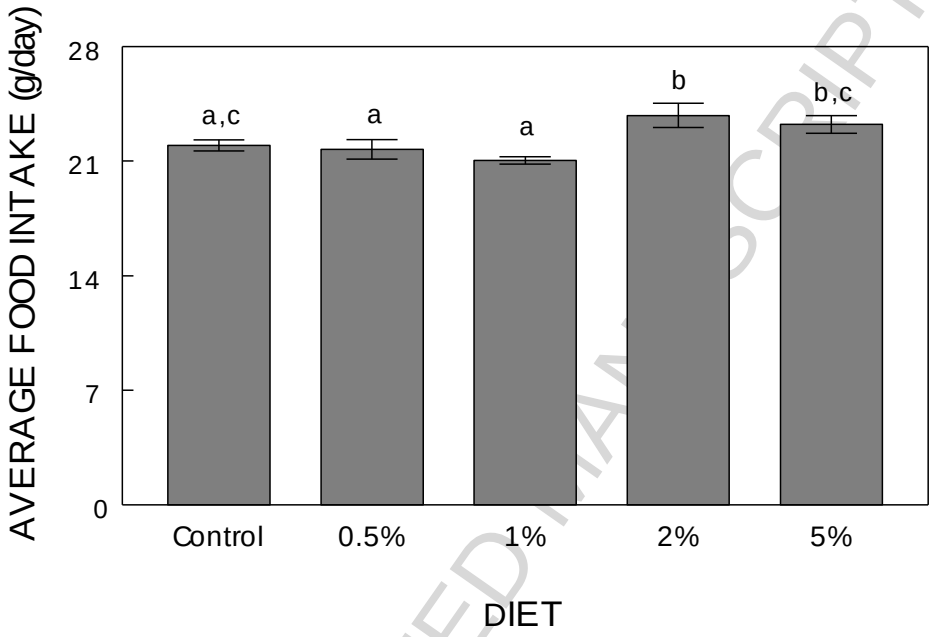
Figure 1

A



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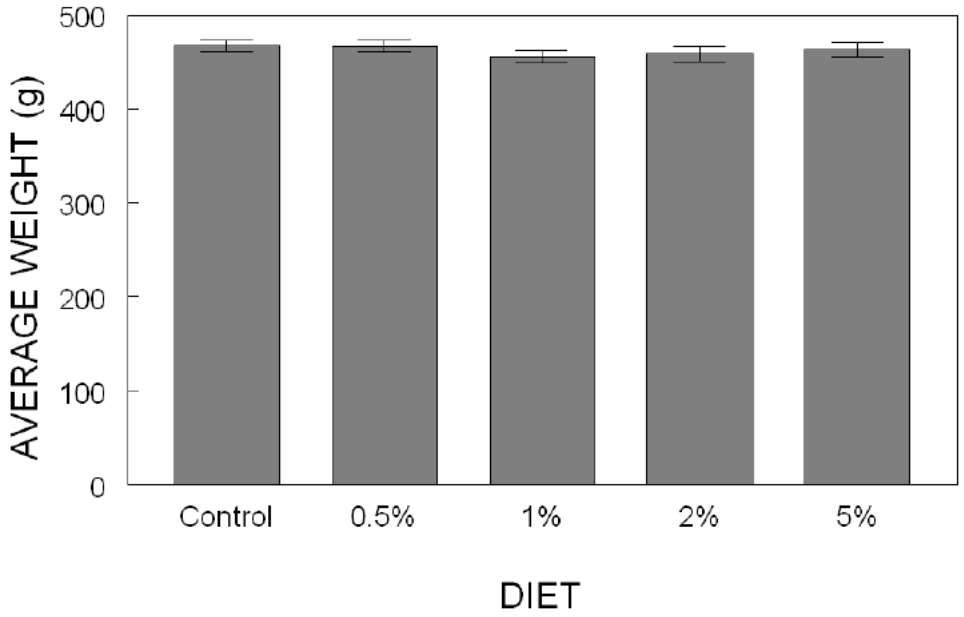

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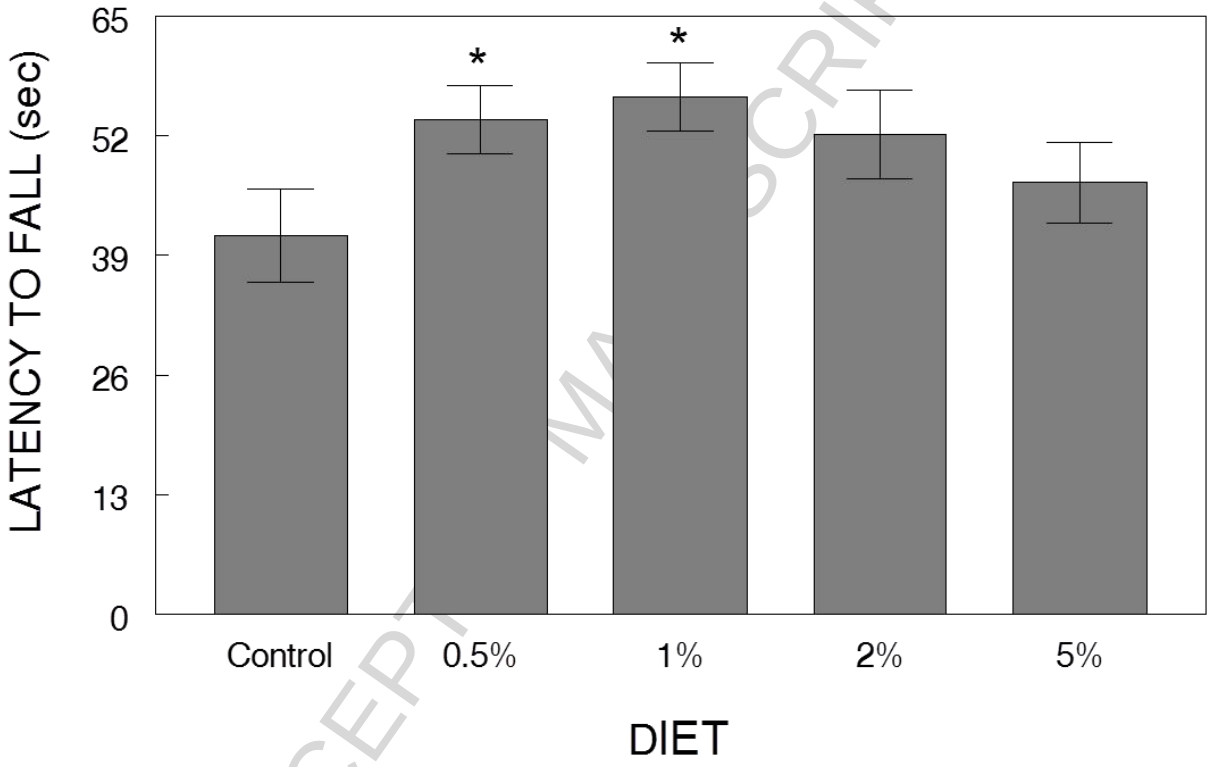

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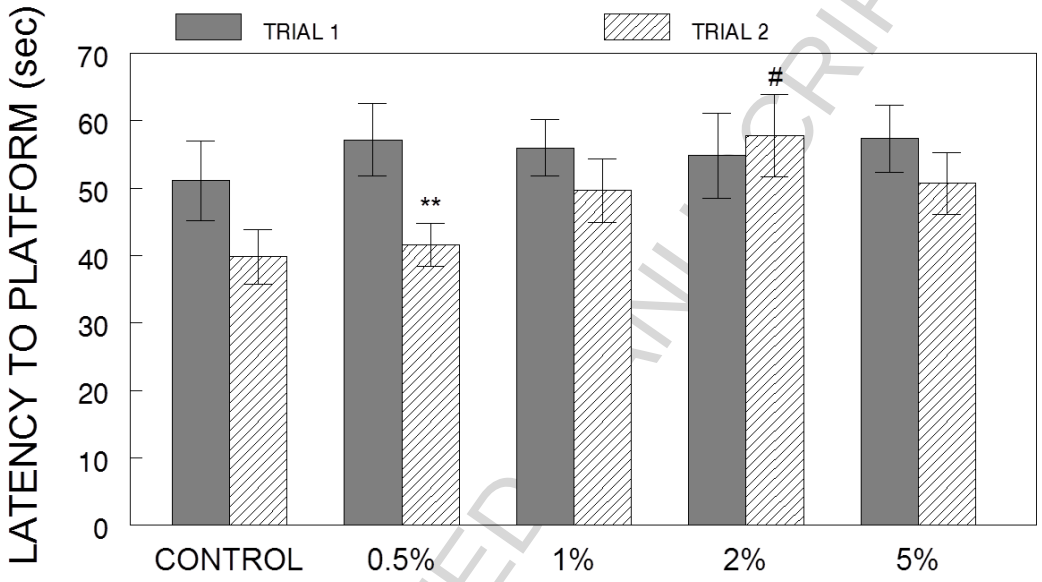
Figure 3

A



B

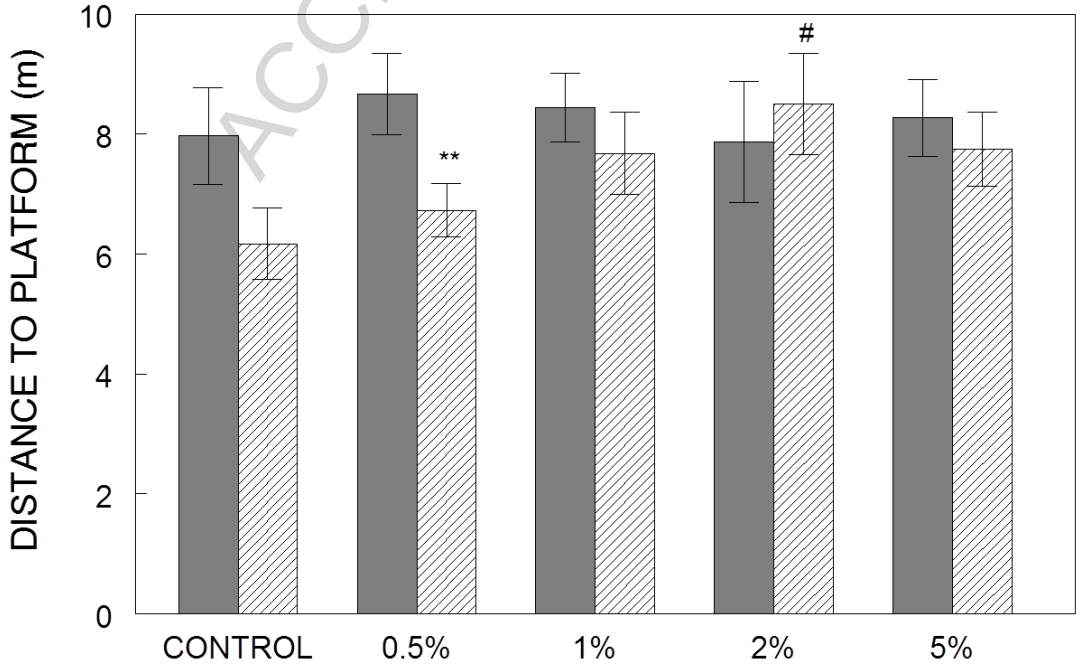


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Table 1. Ingredient composition of the dietfed to rats

Ingredient	**Diet Composition (g/kg)**
Wheat, Hard ground	355.25
Wheat Middlings	100.00
Oats, ground	100.00
Fish Meal, Menhaden, 60%	90.00
Soybean Meal, 48%	50.00
Alfalfa Meal, Dehydrated, 17%	20.00
Corn Gluten Meal, 60%	20.00
Dicalcium Phosphate, 18.5%	15.00
Soybean Oil	15.00
Brewer's Yeast Dried	10.00
Salt, Iodized NaCl	5.00
Calcium Carbonate, 38%	4.75
Vitamin Mix NIH-31[a]	3.50
Mineral Mix, NIH-31[b]	1.50
Corn, yellow ground	110.00
Combination of corn and mushroom powder[c]	100.00
TOTAL	1000.00

Diet is Modified NIH-31[a]Vitamin mix contains (in g/kg vitamin mix): Corn ground (316.24), Choline Chloride (423.50), MSB Complex (135.30), Thiamin mononitrate (25.26), Calcium Panthothenate (24.37), Vitamin B12 (19.05), Vitamin E (17.15), Niacin (12.65), Vitamin A Acetate (10.65), Riboflavin (7.14), Biotin (3.78) Vitamin D3 (2.99) Folic Acid (0.32), Pyridoxine HCl (1.60). [b]Mineral mix contains (in g/kg mineral mix): magnesium oxide (487.10), Calcium Carbonate (219.23), Ferrous Sulfate (146.85), Manganous Oxide (122.40), Copper Sulfate (11.84), Zinc Oxide (10.20), Calcium Iodate (1.74), Cobalt Carbonate (0.64). [c]The amount of corn in the control diet is adjusted accordingly to compensate for the added amount when additional freeze-dried mushrooms were added.

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